graphics market goes, you can be sure you have to spend in excess of Rs 50,000 every few months if you want to have current-generation hardware.

The latest line of graphics cards are multi-GPU cards. These cards have two GPUs on a single card, and essentially offer double the speed of a single GPU card—theoretically. These cards are a boon to those of us with motherboards less than six months old, and only one PCIe slot. Why? Because right now, if you want the best graphics hardware available, it has to be either nVidia-based SLI cards, ATi's CrossFire cards, or a multi-GPU card. There's also the possibility of having two multi-GPU cards in SLI or CrossFire mode, but let's not complicate things!

SLI

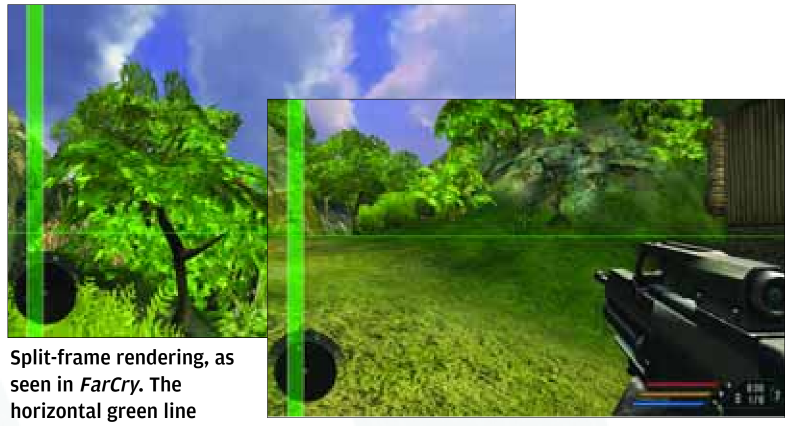
nVidia's SLI (Scalable Link Interface) technology needs a motherboard with two PCIe slots. This technology seems relatively simple: put in two nVidia SLI cards (the cards should be identical; thus you should have two 7800 GTs or two 6800 GTXs, etc.) on to the motherboard. Connect them (like master and slave) via a small daughter card that fits on the top of the cards. Then just connect your monitor to the master, and you're set to go.

Unfortunately, it's not all that simple! The nVidia SLI solution uses technology that was actually created by its erstwhile rival, 3dfx. Back then, SLI stood for Scan Line Interlace. This old technology was used by two graphics cards working in tandem to render alternate lines of a display—one card rendered lines 1, 3, 5, etc., while the other rendered lines 2, 4, 6, etc.

Since nVidia announced its acquisition of 3dfx in 2000, the graphics race was down to only nVidia and ATi. 3dfx's SLI technology resurfaced in 2004 as nVidia's Scalable Link Interface. The major difference was that instead of rendering lines, the new SLI interface would split the 'work' into two, giving each graphics card an equal amount.

This meant that both cards take the same time to finish rendering their allotted work, and this increases performance. Although nVidia had improved the technology earlier, they still had to wait for PCIe to enter the end-user segment—traditional PCI was way too slow for modern GPUs, and AGP didn't allow for dual graphics cards.

nVidia's SLi technology allows two cards to be connected in parallel



Split-frame rendering, as seen in *FarCry*. The horizontal green line distinguishes between the rendering done by each card

How It Works

SLI works in two major ways. The first is Alternating Frame Rendering. Here, each card is given an alternate frame to render by the graphics driver. After the slave card renders its frames, it passes them to the master card, which then adds them alternatively to its own output.

Most games benefit from this form of rendering, as it is easier for the driver to send alternate frames to different cards. Because each frame is rendered on a different card, there is less geometry data that needs to be passed on to each card by the driver. That is, since each card is rendering one frame of a whole scene, each card calculates geometry individually, thereby doubling the geometry output. Most benchmarking software, such as 3DMark, show higher geometry scores because of this.

The second way SLI works is via Split Frame Rendering. This type of rendering divides each frame into work data, and evenly distributes this 'work' between the two graphics cards. This is the technique used to render games such as *FarCry* in SLI mode. Let's take an example of a single frame to better explain how split frame rendering works.

A common scene in *FarCry* is a jungle scene, where, close up, you have lots of plants and foliage, and there's the beach visible on the horizon. As usual, there's the sky above, and there's beautiful rendering of the water in the distance.

Here, the lighter element of the scene is the sky, which needs little or no geometry drawing, and is mainly a simple background. The water (*FarCry*'s speciality), the trees and bushes, however, take a lot of detailing, and thus more geometry and pixel shading. What happens with split frame rendering is that the frame is split into equal processing parts by the driver and then passed on to the cards. Thus, one card gets much more of the frame to render than the other—say, all of the sky and only a part of the trees. The other card gets much less of the frame—the (glorious looking) water and some of the foliage.

The driver calculates the approximate time it will take to render the scene, splits the data in two equal workloads, and then distributes it to the cards. This is always an approximate calculation, because no matter how good the driver is, it cannot predict the exact outcomes of the rendering times. The aim, however, is to get both cards to finish rending their parts in the same